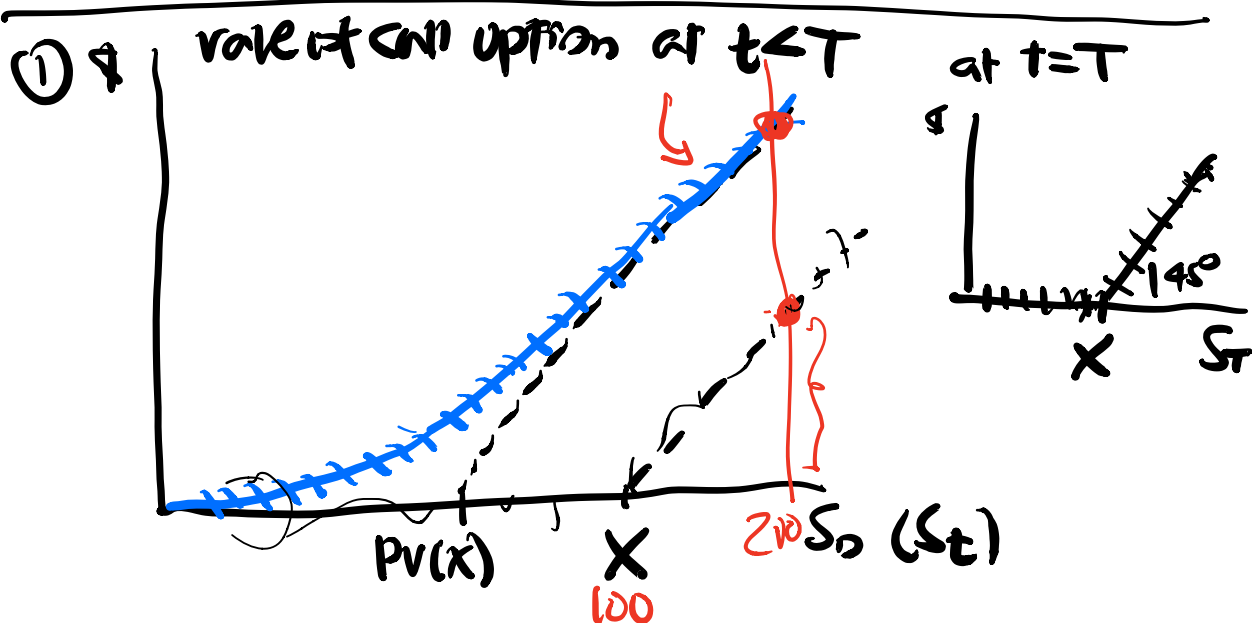
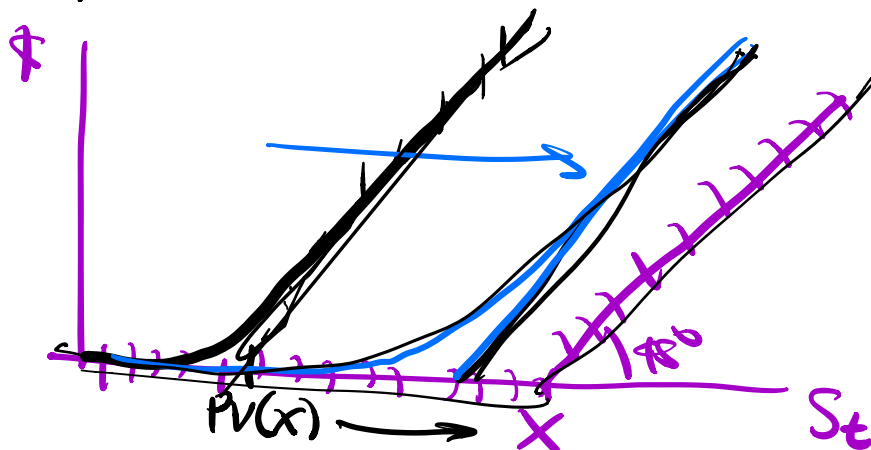


Econ 1710 #12 Friday (Wed) QTA 1pm
 Ch21 option pricing **MONDAY 2-3pm group OH**
 "Prof Kuo"
TUES FINAL EXAM

- ① value of call option (Fig 21-1)
and compared to put option
- ② find stock option pricing - HW #10 Q6 (a)
(b)

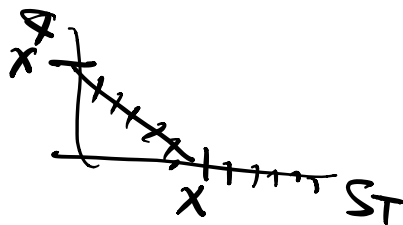


- ① probability is "smoothed out"
- ② approach $PV(X)$ not X



Compare to put option.

- for a call option
the right to exercise early (at $t < T$)
is worthless $\rightarrow$ American Call = European call
- for put option
there is value to early exercise

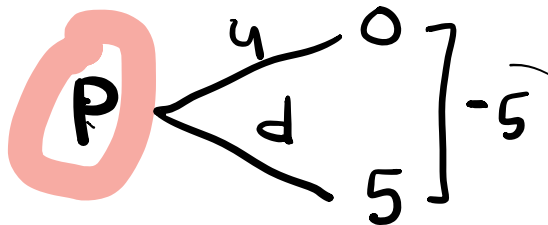
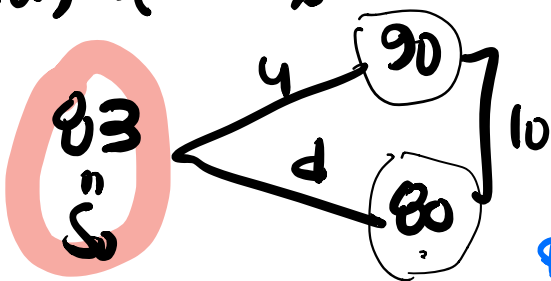


$\Rightarrow$ value of American put $>$ European put

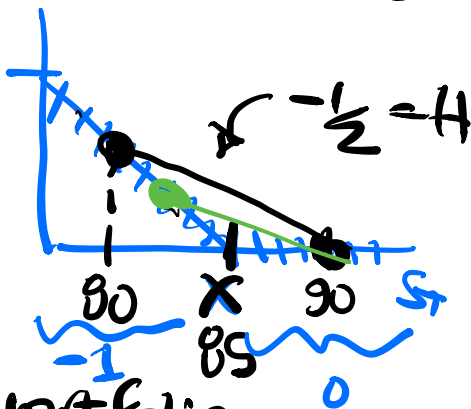
② two state option pricing Hw10 Q26

(a) $r_f = 3\%$

$X = \$85$



$$\Rightarrow H = \frac{\Delta \text{put option}}{\Delta \text{stock value}}$$
$$= \frac{-5}{10} = -\frac{1}{2}$$



⇒ perfectly hedged portfolio
hold 1 share and ~~HOLD~~ 2 puts

negative Δ
means SAME DIRECTION.

hold 1 share hold 2 puts	$\frac{u}{90}$ $\frac{D}{80}$ $\frac{0}{10}$ $\frac{90}{90}$	$\frac{u}{80}$ $\frac{D}{10}$ $\frac{90}{90}$	short 1 share write 2 puts	$\frac{u}{(90)}$ $\frac{D}{(80)}$ $\frac{0}{(10)}$ $\frac{(90)}{(90)}$
-----------------------------	---	---	-------------------------------	---

Then
cost of portfolio today = PV (pay off of portfolio in $T=1$)

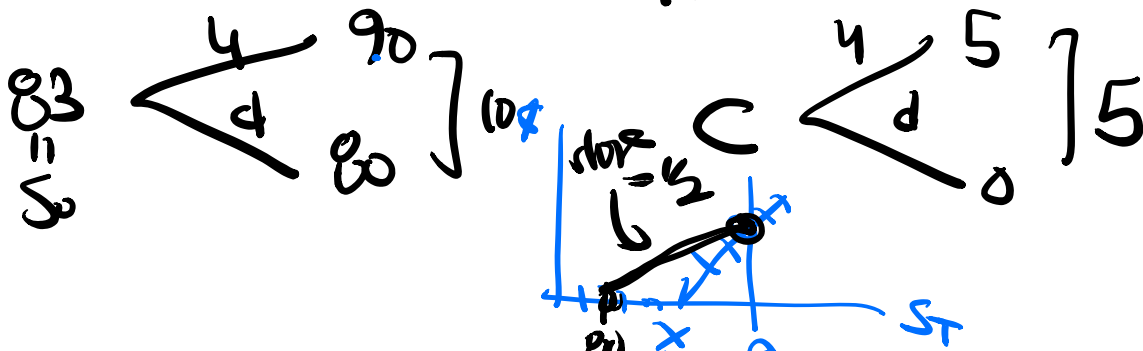
$$-S_0 - 2P = \frac{-90}{1.03}$$

$$P = 2.19$$

$$S_0 + 2P = \frac{90}{1.03} = 87.38$$

$$\begin{aligned} 83 &\rightarrow 2P = \$4.38 \\ P &= \$2.19 \end{aligned}$$

(b) call option instead? $X = \$85$



$$H = \frac{5}{10} = \frac{1}{2}$$

~ 17 85 30

⇒ hold 1 share + write 2 calls

hold 1 share	$\frac{1}{2}$	$\frac{D}{2}$
write 2 calls	$\frac{90}{-10}$	$\frac{80}{0}$
	$\frac{80}{80}$	$\frac{80}{80}$

↑

$$S_0 - 2C = \frac{80}{1.03} = 77.67$$

$$\Rightarrow 2C = 83 - 77.67$$

$$C = \$2.67$$

put call parity

$$C + \text{X} = S_0 + P$$

$$C = S_0 + P - \text{X}$$

$$= 83 + 2.19 - 82.5$$

$$= \$2.67$$